A sharp threshold for three-colour targets from four-colour cubes

Jason Davies

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Abstract

Any collection of 29 cubes, each displaying all four colours of a fixed palette, contains eight that form a $2 \times 2 \times 2$ cube with monochromatic faces using exactly three colours. There is a collection of 28 cubes with no such solution. This answers the threshold question in Problem 2 of Berkove, Eickemeyer and Schulman (2018). Moreover, there are exactly two labelled 28-cube obstructions, exchanged by reflection. The proof uses their reduction by corner containment and elementary counting. The threshold and classification are formalised in Lean.

AI transparency statement. The research argument, computer code, Lean formalisation and manuscript were developed using AI under the author's direction. The declaration at the end gives the scope of this work and its review status.

1 Introduction

Berkove, Eickemeyer and Schulman [1, Problem 2] ask for the least number of four-colour cubes that guarantees a subset of eight forming a three-colour corner solution: a $2 \times 2 \times 2$ cube whose exterior faces are monochromatic and use exactly three colours. Here every source cube uses all four colours of a fixed palette, and repeated varieties are allowed.

We prove that the least number is 29 and classify all collections of 28 cubes that fail. There are exactly two such collections when colour labels are fixed, and reflection exchanges them. Each consists of seven copies of each of four varieties; explicit face tuples are given in Theorem 3. The proof first classifies the possible 28-cube obstructions by corner containment and counting, then deduces the upper bound and verifies that both obstructions fail.

2 Model and geometric facts

Colours are fixed, varieties are identified by proper rotations, and reflections are not allowed as moves. Collections are multisets: repeated varieties are allowed. Thus "labelled" refers to the colour labels, not to labels on individual copies of a cube. Only exterior faces must match the target; internal faces are unrestricted.

A corner is a cyclically ordered triple. Read its face colours counterclockwise as viewed from outside along the corner's body diagonal; equivalently, order the three outward face normals to form a positively oriented basis. Thus abc and acb are distinct when the three colours differ. A source cube fits a target corner by a proper rotation if and only if it has the same cyclic corner type: the cyclic alignments of two corner frames are exactly those realised by proper rotations of the cube. A cube supplies one corner position, regardless of how many suitable corners it has.

Write $Q(a; b, c)$ for a target whose b and c faces are opposite and whose other four faces have colour a. Its corner multiset is $4aab + 4aac$. Write $R(a; b, c)$ for the analogous target with b and c adjacent; its corners are

$$2aaa + 2aab + 2aac + abc + acb.$$

Finally, a target with opposite pairs aa, bb, cc has four corners of each orientation on $\{a, b, c\}$. All three targets use exactly three colours.

If every corner type of a variety v occurs on a variety w, replacing a copy of w by v cannot create a solution that was not possible before: each small cube is used at just one corner. This is the reduction of [1, Section 2]. For clarity, Table 1 gives the entire reduction needed here. A face tuple is read in the order $+x, -x, +y, -y, +z, -z$. The rows represent the seven orbits under colour permutations, with rotations already identified. The three non-retained representatives all contain the corner types of 001123. Colour permutations transport these containments to every member of their orbits.

Representative	Labelled varieties in orbit	Replacement or retained type
000123	12	001123
001123	6	checkerboard C
001213	12	001123
010123	6	001123
010203	8	triple-corner T
010213	12	pair type P
010223	12	pair type P

Table 1: The 68 varieties reduce to 38. The orbits can be obtained by placing the repeated colours: either a triple, or two pairs.

The retained types are precisely the minimal varieties in [1, Table 8]. We use the following explicit description.

- The eight T varieties have a monochromatic corner aaa, and also aab for each of the other three colours b. Each belongs to the bucket of its repeated colour a.
- The 24 P varieties belong to two buckets, one for each repeated colour. In bucket a, such a cube has exactly two of the three types aab with $b \neq a$.
- The six checkerboard varieties have two repeated colours on opposite pairs of faces and two singleton colours on the remaining opposite pair. A checkerboard whose repeated colours are a, b has both orientations on the triples $\{a, b, c\}$ and $\{a, b, d\}$. It belongs to no bucket.

The 32 ordinary varieties (T and P) split into two classes of 16. Every member of one class has the same four trichromatic corner types, one orientation for each three-element subset of the palette; every member of the other class has the four reversed orientations. Each class contains exactly one T variety for each repeated colour. For our coordinate convention the first four types can be taken as

$$\Omega = \{012, 031, 023, 132\}.$$

Table 2 lists the corner types of the four retained representatives. These sets and their colour permutations give the stated incidences and orientation classes. A separate geometric checker verifies both tables and all the stated incidences using physical rotations.

Lemma 1 (Bucket bound)**.** *Eight reduced cubes in a single bucket can build a three-colour target. Consequently, a reduced collection with no such solution has at most seven cubes in each bucket.*

Proof. For bucket a, count the cubes supporting each of the three types aab. Each cube supports at least two types, giving at least 16 incidences. At least two types, say aab and aac, each have support at least four: otherwise the total would be at most $8+3+3 = 14$. Every cube supports at least one of these two types. Thus eight cubes can be matched to four aab and four aac positions, constructing $Q(a; b, c)$. The matching conditions are just support at least four for each type and at least eight for their union. □

Representative	Corner-type set
001123	$\{012, 013, 021, 031\}$
010203	$\{000, 001, 002, 003, 012, 023, 031, 132\}$
010213	$\{001, 003, 011, 012, 023, 031, 112, 132\}$
010223	$\{002, 003, 012, 022, 023, 031, 122, 132\}$

Table 2: Corner types of the retained representatives, with cyclic triples written in lexicographically least form. Multiplicities are omitted: each source cube supplies only one target position.

3 The extremal collections

Theorem 1. *A collection of 28 four-colour cubes without a three-colour corner solution consists of seven copies of each of the four T varieties in one ordinary orientation class.*

Proof. Reduce the collection as above and suppose that no solution exists. Let T, P, C be the numbers of triple-corner, pair-type and checkerboard cubes. Let $A \geq B$ be the sizes of the two ordinary orientation classes, so $A + B = T + P$ and $T + P + C = 28$.

For a three-element colour set E, let z_E count checkerboards supporting its two orientations. If $z_E \geq 8$, those eight checkerboards build the target with three monochromatic opposite pairs. Thus $z_E \leq 7$ for each E, and, since each checkerboard supports two triples, $2C \leq 28$. It follows that $A + B \geq 14$, so $A \geq 7$. If $B + z_E \geq 4$, four cubes from the smaller class and the eligible checkerboards supply one orientation, and four unused cubes from the larger class supply the other. Hence $B + z_E \leq 3$ for every E. Summing gives

$$C + 2B \leq 6, \qquad C \leq 6, \qquad A = 28 - C - B \geq 22. \tag{1}$$

Let t_a and p_a count T and P cubes in bucket a. By Lemma 1, $t_a + p_a \leq 7$. Summing over the four colours gives

$$T + 2P \leq 28, \qquad C \geq P. \tag{2}$$

Suppose first that a checkerboard is present. For every colour a it supports some triple $\{a, b, c\}$. If $t_a \geq 6$, six T cubes supply $2aaa + 2aab + 2aac$, the checkerboard supplies one trichromatic orientation, and an unused cube of the larger ordinary class supplies the other. There is such an unused cube because $A \geq 22$. This constructs $R(a; b, c)$, a contradiction. Therefore

$$t_a \leq 5, \quad T \leq 20, \quad P + C = 28 - T \geq 8. \tag{3}$$

Together with (1) and (2), this implies

$$4 \leq C \leq 6. \tag{4}$$

We claim that $t_a \geq 4$ forces $t_a + p_a \leq 5$. For any two colours a, b, some checkerboard supports a triple containing both: otherwise every checkerboard would have the complementary pair of repeated colours, and one of the corresponding z_E would equal $C \geq 4$, contradicting $z_E \leq 3$. If $t_a = 5$ and $p_a \geq 1$, choose an aab corner on a pair-type cube and a checkerboard supporting $\{a, b, c\}$. The pair-type cube and five T cubes supply the six non-trichromatic corners of $R(a; b, c)$. If $t_a = 4$ and $p_a \geq 2$, two pair-type cubes have a common third colour b, since each supports two of the three possibilities. These two cubes supply $2aab$, and four T cubes supply $2aaa + 2aac$, where a checkerboard again supports $\{a, b, c\}$. In either case that checkerboard and an unused cube of the larger ordinary class fill the last two corners. This proves the claim.

For $t_a \leq 3$, the bucket bound gives $t_a + p_a \leq 7 \leq 10 - t_a$. For $4 \leq t_a \leq 5$, the claim gives $t_a + p_a \leq 5 \leq 10 - t_a$. Summing over all four colours therefore gives

$$T + 2P \leq 40 - T, \qquad T + P \leq 20.$$

But $T + P = 28 - C \geq 22$, a contradiction. Thus $C = 0$, and (2) then forces $P = 0$. All cubes are of type T, and each $t_a = 7$.

If both ordinary orientation classes occur, take a cube from the smaller class, with repeated colour b, and choose $a \neq b$. Six T cubes of repeated colour a supply the six non-trichromatic positions of $R(a; c, d)$, for any distinct $c, d \neq a$. The chosen smaller-class cube and an unused larger-class cube supply the two remaining orientations. This is again a solution. All 28 cubes must therefore belong to one class.

Finally, this conclusion holds for the original collection, not just its reduction. Every non-retained variety in Table 1 is replaced by a checkerboard; since the reduced collection contains none, no such replacement could have occurred. □

Theorem 2. *Every collection of 29 four-colour cubes contains a three-colour corner solution.*

Proof. Otherwise every 28-cube subcollection would have the form in Theorem 1. Take one such subcollection S and let v be the remaining cube. Remove from S a cube of a variety different from v, and insert v. The resulting 28 cubes are not seven copies of each of the four varieties in the original class. They cannot belong entirely to the other class either, since 27 cubes of S remain. This contradicts Theorem 1. □

4 Sharpness

Lemma 2. *In a cube using exactly three colours, the two orientations of a trichromatic corner occur equally often. If no trichromatic corner occurs, the cube is $Q(a; b, c)$ for some distinct a, b, c.*

Proof. Colour the vertices of the dual octahedron by their corresponding face colours and orient its triangular faces consistently. Map every directed edge between two distinct colours to the corresponding directed edge of an abstract triangle on a, b, c, and an equal-colour edge to zero. The sum of the triangular boundaries is zero, since each octahedron edge occurs twice with opposite orientations. A triangle with a repeated colour contributes zero; a trichromatic triangle contributes either $[a, b] + [b, c] + [c, a]$ or its negative. Their numbers are therefore equal.

For the second assertion, if a colour c occurs on only one face, its four neighbours must all have the same colour a: a change from a to b around this face would produce a trichromatic corner. The opposite face is then the sole b face. Otherwise each colour occurs twice. If the two c faces are opposite, the four neighbours of either contain both a and b, forcing a change. If they are adjacent, the other three neighbours form a path containing both a and b, again forcing a change. Both cases produce a trichromatic corner. □

Theorem 3. *The threshold 29 in Theorem 2 is best possible. Exactly two labelled 28-cube collections fail, and they are mirror images.*

Proof. Take seven copies of each of the following four face tuples:

$$010203, \qquad 011213, \qquad 021223, \qquad 031323.$$

These are T cubes with repeated colours $0, 1, 2, 3$, respectively, all in the same ordinary orientation class. A target with any trichromatic corner requires both orientations by Lemma 2, whereas the collection supplies only one. A target without one is $Q(a; b, c)$; every one of its eight corners has colour a twice, so only the seven T cubes of repeated colour a are usable. No three-colour target is possible. Reflection gives the analogous example in the other orientation class. Theorem 1 excludes all other collections. □

5 Formal verification and computation

The threshold and the full classification have been formalised in Lean 4.35.0 [3] with mathlib v4.35.0 [4]. The formal model assigns arbitrary natural-number multiplicities to the 68 four-colour varieties. A physical solution selects eight distinct copies and specifies a proper rotation matching the three exposed faces at each corner of a target. Separate kernel checks establish coverage of all four-colour source and three-colour target face assignments and uniqueness of source varieties up to rotation. A general lemma, valid for any palette and allowing repeated colours, identifies cyclic corner matching with physical face matching by a proper rotation. Finite checks connect the stored corner codes to this lemma.

The formal upper bound follows the counting argument, using finite Hall certificates for the sufficient constructions. The lower bound checks a Hall deficiency for each of the 120 targets and each extremal collection; reflection exchanges the two collections. Thus the formal proof does not enumerate all collections. Its axiom audit reports only Lean's standard `propext`, `Classical.choice`, and `Quot.sound`; it uses neither proof holes nor `native_decide`. The accompanying repository[1] contains the source and reproduction commands. It reuses the proper-rotation and outward-face geometry from the earlier MacMahon formalisation, pinned to commit `3f854795a77cd7c902e4c5a1a5da3322f9cd44fa`. No classification result from that project is used.

The proof does not depend on optimisation. A Hall-obstruction model first found the 28-cube example. After the bucket reduction, the compiled multicore CP-SAT backend also reported the maximum obstruction size as 28. That solver status is a computational check, not a separately verified optimality certificate.

Two further feasibility runs reported infeasibility for a 29-cube obstruction and for a 28-cube obstruction other than the two specified collections. Both use the reduced source family and the bucket bounds.

The accompanying script `check_three_colour_threshold.py` independently generates the 24 rotations from quarter-turns, verifies the 38-type reduction and all corner incidences, and checks the lower example against all 120 three-colour targets on the four-colour palette by bipartite matching and explicit Hall deficiencies. It also checks all 68 one-cube extensions of the lower example and 1,000 sampled collections of size 29. These samples do not establish the upper bound; the written proof does.

The result answers the threshold question in Problem 2 of [1]; whether it is the first answer remains unresolved. An expanded, bounded literature search on November 1, 2026 found no later answer, but does not establish priority. In particular, Rosenfeld's October 2026 article [2] was identified but could not be read, so its relevance to this question has not been determined. A separate AI review found no mathematical errors in the argument or physical model. These automated checks are not independent human peer review.

Declaration of AI-generated research and review

GPT-6-Astra (OpenAI), used through Codex, developed the research argument, computational search, Lean formalisation and manuscript under the author's direction. A separate AI reviewer checked the argument and physical model. The human author initiated and directed the research and accepts responsibility for the content of this paper and the accuracy of this disclosure. The paper has not received independent human peer review; formal verification does not establish publication priority.

[1] https://github.com/jasondavies/macmahon-four-to-three

References

[1] E. Berkove, A. Eickemeyer and M. Schulman, *The 4-Color Cubes Puzzle*, Integers **18** (2018), A68. Journal PDF.

[2] M. Rosenfeld, *MacMahon Cubes and Moshe's Insanity, Puzzle 1*, Geombinatorics Quarterly **36**(2) (October 2026), p. 78 ff. Publisher listing (full text not examined).

[3] L. de Moura and S. Ullrich, *The Lean 4 Theorem Prover and Programming Language*, in Automated Deduction—CADE 28, LNCS **12699**, Springer (2021), 625–635. doi:10.1007/978-3-030-79876-5_37.

[4] The mathlib Community, *The Lean Mathematical Library*, in Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM (2020), 367–381. doi:10.1145/3372885.3373824. arXiv:1910.09336.